Digital Power Module

CANBUS Communication Protocol

Beian New Energy Co., Ltd.

Beian New Energy Co., Ltd.

1 Scope

This communication protocol adopts CAN specification version 2.0B, 29-bit identifier, and extended frame.

2 Range

This CANbus communication protocol describes the external CAN communication protocol of Beian New Energy Co., Ltd. It is applicable to the 10kW series, 12kW series, 20kW series, 30kW series charging modules and laser power supplies and other power modules produced by Beian New Energy Co., Ltd.

3 Physical communication interface

Canbus. Baud rate 125kbps/s

4 Frame Structure

29-bit identifier as below:

28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte3					Byte2								Byte1								Byte0							
Command ID_H																						Module Address						

Beian New Energy Co., Ltd.

Command Define 1:

Functions	CAN ID	Module Address	Data Length	Byte							
				0	1	2	3	4	5	6	7
Output Voltage/current Setting	(0x1907,C080)>>7	0~60	8	CMD=0	Current(mA)*1000			Voltage(mV)*1000			
Module Return	(0x1807,C080)>>7	1~60	8	CMD=0	NZ,NZ Current Success Voltage Success						
					0,NZ Current Fail Voltage Success						
					NZ,0 Current Success Voltage Fail						
					0,0 Current Fail Voltage Fail						
Read output voltage/current	(0x1907,C080)>>7	1~60	8	CMD=1	/	/	/	/	/	/	/
Module return	(0x1807,C080)>>7	1~60	8	CMD=1		Current (A*10)		Voltage (V*10)		Status 1	Status 0
Power On/OFF Setting	(0x1907,C080)>>7	0~60	8	CMD=2	/	/	/	/	/	/	0x55 Power ON 0xAA Power OFF
Module return	(0x1807,C080)>>7	1~60	8	CMD=2	NZ,NZ: Success 0,0:Fail						

Note:

Address 0 is a broadcast command, which is executed for all modules on the CANbus. The broadcast command module does not respond. Addresses 1 to 60 represent module addresses, and the corresponding module addresses will reply with data.

NZ represents non-zero data (default 0xFF). The data field is transmitted first with the higher byte and then with the lower byte. If Module 2 does not receive the monitoring command within 2 minutes, it is considered that the communication has been interrupted.

"/" indicates the current reservation, and 0 is filled in.

Beian New Energy Co

Definition of module status flags:

Byte7:Bit	Status0	Panel LED
Bit 7	**1:** Output over low voltage **0:** Normal	**1:** YELLOW LED ON
Bit 6	**1:** Output over high voltage **0:** Normal	**1:** RED LED ON
Bit 5	**1:** Input over low voltage **0:** Normal	**1:** RED LED ON
Bit 4	**1:** Input over high voltage **0:** Normal	**1:** RED LED ON
Bit 3	**1:** Fan fail **0:** Fan normal	**1:** RED LED FLASHING
Bit 2	**1:** Module output constant current **0:** Module output constant voltage Bit2 is status, not fail status, please don't make it into fail group.	**1:** YELLOW LED ON
Bit 1	**1:** Module fail **0:**Normal	**1:** RED LED ON
Bit 0	**1:** Module power OFF **0:** Module power ON	/

Byte6:Bit	Status1	Panel LED
Bit 7		
Bit 6		
Bit 5		
Bit 4		
Bit 3		
Bit 2	**1:**Setting the module power OFF **0:** Setting the module power ON	**1:** GREEN LED FLASHING **0:** GREEN LED ON
Bit 1	**1:** Over temperature protection **0:**Normal	**1:** RED LED ON
Bit 0	**1:** Over current protection **0:** Normal	**1:** RED LED ON

Note:

For the above alarms, except when the panel indicator light is on red, the module will automatically shut down for protection. For other modules, normal output will be maintained.

Beian New Energy Co., Ltd.

Command Define 1:

Functions	CAN ID	Module Address	Data Length	Byte							
				0	1	2	3	4	5	6	7
Read Setting Voltage Value	(0x1901,0080)>>7	1~60	0	Empty (Reading the voltage setting in the module, to confirm the value is right)							
Module return	(0x1801,0080)>>7	1~60	8							Voltage Setting (V*10)	
Read Setting Current Value	(0x1901,0880)>>7	1~60	0	Empty (Reading the current setting in the module, to confirm the value is right)							
Module return	(0x1801,0880)>>7	1~60	8	Max Current Setting(A*10)							
Read Input Voltage Value	(0x1907,A080)>>7	1~60	8	0x31	/	/	/	/	/	/	/
Module return	(0x1807,A080)>>7	1~60	8	0x31		Vab/32		Vbc/32		Vca/32	
						Signal phase V/32					
						DC input V/32					
Read environment temperature	(0x1900,8080)>>7	1~60	0	Empty							
Module return	(0x1800,8080)>>7	1~60	8					Temperature (℃*10)			
Module uniform current control	(0x19C2,1880)>>7	0	6	/	/	/	0x55 POWER OFF				
							0xAA POWER ON				
Internal uniform current in the module	(0x1807,8080)>>7	1~60	8	The module automatically reports the average current data every 400ms, and users do not need to pay attention to it.							
Module address search	(0x19C2,2880)>>7	1~60	8				0x55 GREEN LED FLASHING				

Beijing Energy Co., Ltd.

5 Frame Example.

1. Broadcast setting the module ON/OFF

Send ID	1907C080	**Data**	2	0	0	0	0	0	0	55
Return ID		**Data**								

Broadcast command no reply

2. Setting the module 1 POWER ON/OFF

Send ID	1907C081	**Data**	2	0	0	0	0	0	0	55
Return ID	1807C081	**Data**	2	FF	0	0	0	0	0	0

If the setting is successful, Byte1 will return FF; if it fails, it will return 00.

3. Setting the module 3 output voltage and current

Send ID	1907C083	**Data**	0	01	5F	90	00	01	86	A0
Return ID	1807C083	**Data**	0	FF	0	0	0	0	0	0

Module 3 setting the output voltage 100V, Current 100A
100V = 100*1000 = 100000mV = 0x186A0
90A = 90*1000 =90000mA = 0x15F90
If the setting is successful, Byte1 will return FF; if it fails, it will return 00.

4. Read module 2 status information

Send ID	1907C082	**Data**	1	0	0	0	0	0	0	0
Return ID	1807C082	**Data**	1	0	01	17	03	E8	0	0

Output Voltage Value: Byte4~Byte5 0x03E8 = 1000/10 = 100V
Output Current Value: Byte2~Byte3 0x0117 = 279/10 = 27.9A
Output Status: Byte6~Byte7 0x00 0x00 Normal, No warning.

5. Read module 1 output voltage

Beian New Energy Co., Ltd.

Send ID	19010081	**Data**								
Return ID	18010081	**Data**	1	0	18	36	0	1E	03	84

Module 1 output voltage setting value: Byte6~Byte7 0x0384 = 900/10 = 90V

6. **Read module 5 output current(Limit current)**

Send ID	19010885	**Data**								
Return ID	18010885	**Data**	01	3D	0	0	0	0	40	2C

Module 5 output current setting value: Byte0~Byte1 0x013D = 317/10 =31.7A

7. **Read module 1 input voltage**

Send ID	1907A081	**Data**	31	0	0	0	0	0	0	0
Return ID	1807A081	**Data**	31	01	2F	1E	2F	2C	2F	26

Module 1 input voltage

Vab: Byte2~Byte3 0x2F1E = 12062/32 =376.9V

Vbc: Byte4~Byte5 0x2F2C = 12076/32 =377.4V

Vca: Byte6~Byte7 0x2F26 = 12070/32 =377.2V

8. **Read module 1 environment temperature**

Send ID	19008081	**Data**								
Return ID	18008081	**Data**	0	1F	0B	B5	01	52	03	A1

Environment temperature: Byte4~Byte5 0x0152 = 338/10 =33.8°C

9. **Disable the module's automatic current balance**

Send ID	19C21880	**Data**	0	0	0	55	00	00		
Return ID		**Data**								

Byte4 = 0x55 Current balance OFF, 0xAA Current balance ON;

10. **Module address search**

Send ID	19C22882	**Data**	0	0	0	55	0	0	0	0

Beian New Energy Co., Ltd.

Return ID		Data								

Search the Module 2. The green light of Module 2 will flash for 10 seconds.

6 Cautions

1. During the soft-start process, the module cannot be directly loaded. It can only be loaded after the output voltage of the module is established (output is stable). This can be determined by checking the module status flags (refer to Table 6-1 and Figure 6-1).

Byte7:Bit	Status0	Panel LED
Bit 7	**1:** Output over low voltage **0:** Normal	**1:** YELLOW LED ON
Bit 6	**1:** Output over high voltage **0:** Normal	**1:** RED LED ON
Bit 5	**1:** Input over low voltage **0:** Normal	**1:** RED LED ON
Bit 4	**1:** Input over high voltage **0:** Normal	**1:** RED LED ON
Bit 3	**1:** Fan fail **0:** Fan normal	**1:** RED LED FLASHING
Bit 2	**1:** Module output constant current **0:** Module output constant voltage Bit2 is status, not fail status, please don't make it into fail group.	**1:** YELLOW LED ON
Bit 1	**1:** Module fail **0:**Normal	**1:** RED LED ON
Bit 0	**1:** Module power OFF **0:** Module power ON	/

Table 6-1

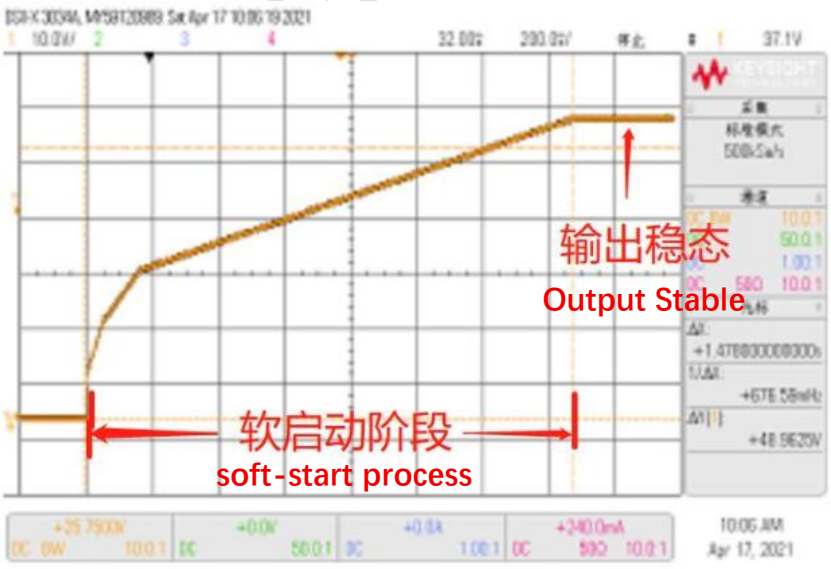


Figure 6-1

2. The output current of the module should be gradually increased in steps. For example, if the required current is 100A, the initial current setting can be set at 10A, and then increase by 10% of the load current, i.e., in increments of 10A, 20A, ... , 90A, 100A. It is recommended that the current ramp-up time be ⩾ 1 second.